Journal of
Theoretical and Applied Electronic Commerce Research

MDPI

Article

An Analysis of Consumer Purchase Behavior Following Cart Addition in E-Commerce Utilizing Explainable Artificial Intelligence

Ramazan Esmeli [1,*] and Aytac Gokce [2]

[1] Department of Computer Engineering, Engineering Faculty, Van Yuzuncu Yil University, Van 65090, Turkey
[2] National Education of Turkiye, Ankara 06420, Turkey; aytac.gokce@meb.gov.tr
* Correspondence: ramazanesmeli@yyu.edu.tr

Abstract: To optimize personalized offers and reduce cart abandonment, it is essential to understand customer behavior in e-commerce after products are added to the cart. Although purchase prediction models are well researched, session-level changes, including price variations, product category shifts, and geographical context, are less examined concerning their impact on machine learning models for predicting purchase behavior after cart additions. This study incorporates these factors into machine learning models to examine their impacts on predictions using explainable AI techniques. The comprehensive experimental results obtained from two datasets and eight models demonstrate that machine learning algorithms can achieve an F1 score of 89% in predicting purchase behavior following cart additions. This study highlights the significant impact of session-specific factors, like price fluctuations, category transitions, and geographical context, coupled with consumers' previous browsing patterns, on model predictions.

Keywords: cart abandonment; purchase prediction; price changes; product category shifts; geographic context; machine learning techniques

check for updates

Academic Editors: José Luís Mendes Loureiro Abrantes, Natália de Lima Figueiredo, Bruno Morgado Ferreira and Luís F. Martinez

Received: 12 December 2024
Revised: 30 January 2025
Accepted: 10 February 2025
Published: 13 February 2025

Citation: Esmeli, R.; Gokce, A. An Analysis of Consumer Purchase Behavior Following Cart Addition in E-Commerce Utilizing Explainable Artificial Intelligence. *J. Theor. Appl. Electron. Commer. Res.* **2025**, *20*, 28. https://doi.org/10.3390/jtaer20010028

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

The popularity of online shopping has surged considerably, intensifying competition among e-Commerce platforms striving to both attract and retain customers. In this competitive landscape, businesses adopt several strategies to boost visibility and sales. A crucial method is utilizing machine learning models to gain insights into customer behavior during product browsing and cart addition [1,2]. However, enhancing conversion rates presents a challenge, mainly due to frequent shopping cart abandonment. Studies reveal a high rate of abandonment, often caused by unexpected costs, trust issues, or indecision [3–5]. Overcoming these hurdles is essential to improve customer engagement and increase conversions. ML significantly helps in addressing cart abandonment by analyzing extensive behavioral data to identify patterns and predict outcomes [6]. Personalized recommendations through ML, such as collaborative filtering, are effective in increasing conversion rates [7]. Additionally, real-time interventions, such as cart reminders and targeted discounts, prompted by predictive models, can enhance the likelihood of purchasing [6,8,9]. Beyond predictions, ML assists with A/B testing and customer segmentation, allowing platforms to refine user journeys and adopt effective marketing strategies [10,11].

Recent studies focus on predicting purchase outcomes using session logs but overlook critical analyses of post-cart-addition behavior, where high abandonment rates persist despite user intent. To address this gap, we propose a machine learning framework enhanced

with explainable AI (XAI) to clarify decision-making processes after cart additions, using datasets from a UK SaaS e-commerce platform. Our framework employs SHapley Additive exPlanations (SHAP) to interpret model behavior, identify novel predictors (e.g., pricing volatility, category transitions, geographic context), and provide actionable recommendations for e-Commerce practitioners. By evaluating advanced ML models on real-world data, we aim to optimize prediction accuracy while answering the following research questions:

1. How do product category transitions during sessions affect the decision making of prediction models?
2. How do session-level pricing variations, including volatility and relative pricing, impact predictive model decision-making processes?
3. How do geographic characteristics influence the decision-making processes of predictive models?

The structure of this paper is as follows: Section 2 examines related literature, Section 3 introduces the proposed framework, and Section 4 describes the dataset analysis. Section 5 covers the experimental results and their interpretation, while Section 6 highlights the primary findings, contributions, and study implications. Finally, Section 7 concludes by noting the research limitations and proposing future research directions.

2. Related Work

E-commerce businesses face a notable challenge in reducing online shopping cart abandonment to improve conversion rates. Several studies have explored the psychological, behavioral, technological, and environmental factors affecting this behavior. As discussed in Rausch et al. [12], machine learning models were assessed for their ability to forecast cart abandonment using data from a major German retailer. Models such as gradient boost, decision tree, and logistic regression were compared, with the results showing that while gradient boost with regularization had the highest accuracy, decision tree and logistic regression strike a more favorable balance between complexity and performance due to computational practicality. Kukar-Kinney and Close [13] examined psychological influences, showing that shopping carts act as organizational tools. They found that consumers using carts for research are less likely to abandon them if a price drop is expected, adding complexity to consumer behavior models by identifying deterrents like entertainment value and privacy concerns. Rubin et al. [14] investigated the cognitive aspects of consumer behavior, indicating that an abstract mindset promotes purchase completion, emphasizing the benefit of fostering such a mindset to boost conversions.

Padigela and Suguna [15] proposed a recommendation system targeting cart abandonment by employing collaborative filtering. This system uses cosine similarity for customer segmentation and KNN for personalized product recommendations, which they claim curtail abandonment and boost sales. Furthermore, Jiang et al. [16] discovered that displaying cart products in ascending order mitigates abandonment by reducing forgetfulness and choice overload, suggesting that strategic organization by e-retailers can enhance user experience. Additionally, Ref. [4] explored abandonment via a sequential mediation model within the stimulus–organism–response framework, revealing that anticipated lower prices spur checkout hesitation and abandonment, amplified by perceived inconveniences. Hence, improving the checkout process and reducing barriers could effectively decrease abandonment. Recent studies also investigate consumer behavior influences. Zhao et al. [17] examined cart warning pop-ups, finding that they polarize consumer preferences and widen the intention gap due to anticipated regret. Kapoor and Vij [3] distinguished between cart and checkout abandonment, noting that the former is affected by price and shipping policies, while the latter is mainly due to account fatigue, with demographic differences observed in abandonment timing.

Adding to these findings, Li et al. [18] introduced the role of environmental factors in the abandonment of online shopping carts, showing that sunny weather is related to lower abandonment rates, with mood and perceived risk mediating this effect. Their study also found that brand reputation amplifies the positive impact of sunny weather on reducing abandonment, suggesting that environmental conditions should be integrated into e-commerce strategies. Ref. [19] highlighted that transaction and product risks significantly contribute to mobile shopping cart abandonment, with personal or technology risks having less impact. Enhancing information quality and consumer trust can manage these risks. Wang et al. [20] found that fear appeals during COVID-19 increased consideration of physical store purchases despite checkout uncertainty causing higher abandonment. Deep learning shows that engagement data like session time and interaction frequency are valuable for customer purchase predictions [21].

Numerous studies have investigated sophisticated methods for predicting consumer behavior. For instance, traditional probabilistic models have been integrated with advanced machine learning techniques, such as extreme gradient boosting, to enhance the accuracy of predictions with profit-maximizing objectives, demonstrating improved performance in empirical datasets [22]. The study by Ehsani and Hosseini [23] introduced a composite meta-classifier that enhances prediction precision and reduces error rates in purchase forecasts. Meanwhile, another investigation by [24] underscored the importance of incorporating customer characteristics and browsing behavior to achieve superior prediction outcomes. Additionally, Trivedi et al. [25] proposed a novel algorithm for predicting purchase intentions, which addresses data imbalance and employs a two-stage feature selection process. This model, leveraging ensemble classifiers, exhibited enhanced accuracy over traditional classifiers, offering valuable insights for more efficient customer targeting. The research conducted by Wei et al. [26] introduced a model to predict customer preferences by utilizing a confluence of profile data, purchase histories, and browsing records. A Bayesian nonparametric approach to sales prediction was proposed by Wu et al. [27], whose model effectively manages intricate sales trends and distribution shifts while providing explicit and uncertain prediction insights. Moreover, Ref. [28] introduced the RCFF framework, which combines CatBoost and RF algorithms to identify potential buyers, demonstrating superior efficacy in comparison with traditional classifiers. Finally, numerous investigations have concentrated on predicting real-time behavior. The study executed by Chen et al. [29] examined the effect of online actions on the joint purchasing of products, indicating that actions such as adding items to the cart can influence purchasing decisions based on the product's context, brand, and customer reviews. Research by Mikalef et al. [30] employed eye-tracking data to anticipate consumer intentions on social media platforms. Furthermore, Sakthi and Sundar [31] developed a hybrid deep learning network for predicting consumer behavior, noting its superior performance relative to conventional methodologies. Similarly, Esmeli et al. [32] emphasized the relevance of contextual and loyalty attributes in the premature prediction of purchasing behaviors, and Sakar et al. [33] established a framework aimed at predicting purchase intentions in real time.

Literature Gaps and Our Contributions

While there has been significant progress in predicting purchase and cart abandonment behaviors, session-level dynamics remain underexplored. Research often focuses on aggregated customer characteristics or fixed attributes, overlooking temporal and contextual browsing shifts. Specifically, the following are overlooked:

1. Product category dynamics: Limited studies examine the impact of customer movement between product categories during a session on purchase decisions or abandon-

ment risks. These shifts potentially indicate changing interests or confusion, essential for understanding real-time intent.

2. Price dynamics: Research frequently neglects the influence of price alterations or user comparisons during sessions on behavior. Aspects like price drops, promotions, or price-sensitive responses are vital for predicting purchases and cart abandonment.
3. Locational attributes: While demographic and behavioral aspects are well studied, the effect of locational context, including geographical constraints, regional preferences, or device locations, is less explored. These can improve models by adding context that links behavior to external factors.

Exploring these elements provides a refined understanding of session-level behavior. Including such overlooked features in predictive models enhances the accuracy of purchase and abandonment predictions, offering deeper insights for e-commerce platforms.

3. Proposed Framework

To contextualize this study, we align our work with the buyer's decision process as defined by Kotler and Keller [34], p. 195, which outlines five key stages: (1) problem recognition, (2) information search, (3) evaluation of alternatives, (4) purchase decision, and (5) post-purchase behavior. Although stage 2 (information search) is partially reflected in metrics like the number of browsed products, our analysis focuses on stages 3 and 4, where critical decision making occurs. We used e-commerce transaction logs to investigate how behavioral factors, such as the number of item categories explored, price differences between alternatives, geographical disparities (latitude–longitude data), and prior purchase history, interact to predict cart abandonment or purchase completion. By applying ML models, we assess the predictive power of these variables that moderates decision-making dynamics during the evaluation and purchase stages. The analysis of consumer behavior indicators using ML models improves marketing profitability by predicting purchase actions or cart abandonment. Behavioral data reveal patterns in consumer preferences, aiding in these predictions. The framework enhances personalized marketing through navigational data insights, aiding in cart abandonment and purchase predictions. It has three stages: 1. Data Collection and Pre-processing: gather and prepare data for analysis. 2. Feature Engineering: identify consumer behavior traits from session logs that affect cart decisions. 3. Model Training and Evaluation: train ML models and evaluate the ML models' ability to predict abandonment or purchase, guiding marketing tactics. Figure 1 (area "a") shows an application on an e-commerce platform. ML models use extracted features to determine whether users will purchase or abandon, influencing adaptive marketing strategies such as price adjustments or recommendations. This framework aims to enhance customer engagement and decision making.

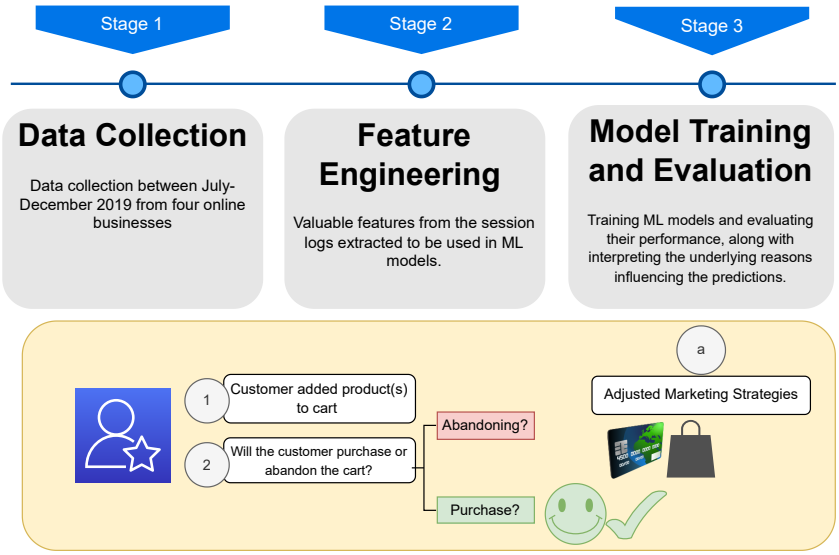
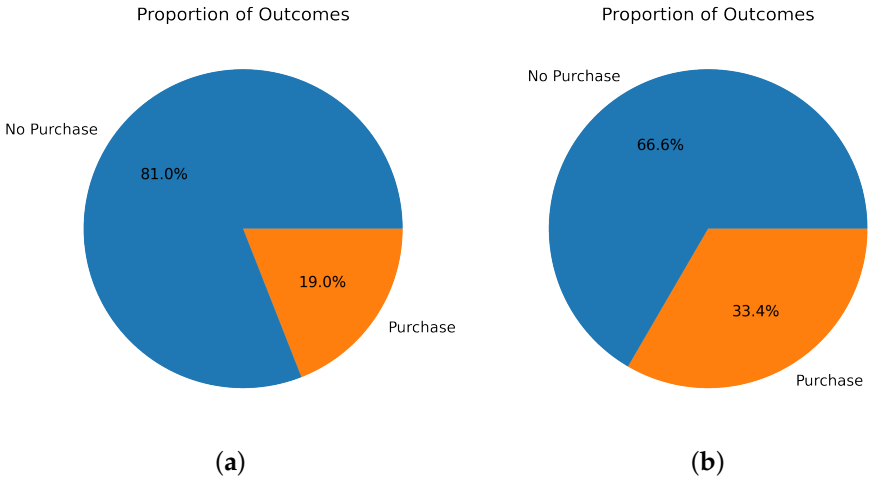


Figure 1. The proposed framework.

3.1. Stage 1: Data Collection

In the first stage of the framework, attributes regarding consumer behavior on the web platform are collected as raw data. The online platforms from which data are collected have a wide range of products, including tools and clothing and protection. In this research, data are collected from two different datasets belonging to two different businesses by a software-as-a-service solutions company based in the UK. A description of each dataset of the number of interactions, consumers, and products is provided in Table 1. In dataset 1 (ds1), approximately 81% of the interactions result in cart abandonment, leaving about 19% of the sessions ending with a purchase. Similarly, in dataset 2 (ds2), the cart abandonment rate is around 67%, while approximately 33% of the interactions end with a purchase (Figure 2).



(**a**) (**b**)

Figure 2. Proportion of outcomes for ds1 and ds2. (**a**) Proportion of outcomes for ds1; (**b**) Proportion of outcomes for ds2.

Later, necessary features such as time, location, browser type, operating system type, and previous purchases and visits are selected from the collected raw data.

Table 1. Summary of dataset statistics.

Dataset	#Interactions	#Sessions	#Items	#Cart	#Purchase
ds1	209,379	72,083	13,571	169,524	39,855
ds2	78,841	26,151	19,367	52,521	26,320

3.2. Stage 2: Feature Engineering

To improve the model's predictive accuracy for cart abandonment, we utilized feature engineering to convert raw session data into structured insights. Below is a summary of the extracted features:

- pages viewed: Total page views by the user in a session, including repeats.
- max price: Highest price of any product viewed during the session.
- Longitude and Latitude: Approximate location of the user.
- #previous sessions: User's prior session count, indicating platform history.
- #cat changes: Number of category switches during the session.
- ff pc: Binary indicator for sessions on a PC.
- hod: Hour of session, based in 24 h time.
- min price: Lowest product price seen in the session.
- products viewed: Number of distinct products viewed.
- unique prices: Count of different prices encountered.
- price difference: Gap between highest and lowest prices viewed.
- dow: Numeric day of the week (1–7) of the session.
- dom: Numeric day of the month (1–31) of session occurrence.
- ff tablet: Binary marker for sessions on a tablet.
- weekend: Binary flag for weekend sessions.
- price change: Binary check for price changes during interaction.

To quantify the price dynamics during a user session, we define the following metrics using the sequence of prices browsed in session i, denoted as $P_i = \{p_1, p_2, \ldots, p_n\}$. We present an example of how we extract this information from a session as follows: Consider a session with prices $P_i = \{\$10, \$15, \$10\}$:

- min price(P_i) = USD 10
- max_price(P_i) = USD 15
- price_difference(P_i) = USD 5
- price_change(P_i) = 1 (prices changed)
- unique_prices(P_i) = 2

3.3. Stage 3: Model Development and Evaluation

The third stage of the framework is for building the ML prediction models for consumers' purchases. In this stage, eight state-of-the-art ML models are trained to evaluate how different models perform with purchase prediction. These eight models are logistic regression (LR) [35], random forest (RF) [36], gradient boosting (GB) [37], k-Nearest Neighbor (KNN) [38], decision tree (DT) [21], Multi-Layer perceptron (MLP) [33], Convolutional Neural Network (CNN) [39], and Long Short-Term Memory (LSTM) [33]. Each model has its own specifications and parameters, which are explained in detail in the Experiments and Results section. Given the significant imbalance in the datasets, we evaluated the performance of each ML model on a balanced version of the datasets.

The models were assessed to identify the superior one in predicting consumer purchase and cart abandonment behaviors. ROC AUC, recall, precision, and F1 score were employed for the evaluation of ML models. Recall indicates the percentage of actual positive cases

accurately recognized by the model, measuring the model's capability to retrieve relevant results, and is computed as in Equation:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Precision shows the proportion of predicted positive instances that were actually correct. It evaluates how accurate positive predictions are and is calculated as in Equation:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

F1 Score shows the harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives. It is calculated as in the following Equation:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

ROC-AUC (Receiver Operating Characteristic–Area Under the Curve) shows the model's ability to distinguish between classes. It represents the probability that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance. ROC-AUC is calculated by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at various classification thresholds, with TPR and FPR defined as follows:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

The area under this curve gives the ROC-AUC score. These scores are calculated based on the 10-fold cross-validation results.

4. Dataset Analysis

In order to understand the dataset characteristics better, we conducted several analyses on the datasets. First, we started with a daily and hourly outcome analysis for ds1 and ds2. Figures 3 and 4 reveal that the daily and hourly browsing patterns for ds1 and ds2 are similar, highlighting user behavior. Both datasets show steady browsing throughout the week, with more frequent "b" (added to cart) actions. Although less common, "t" (transaction) outcomes show a small uptick on weekdays, implying consistent browsing but increased weekday transactions. Both datasets exhibit a clear daily pattern: "b" activities peak overnight and in the early morning (midnight to 6 a.m.) and drop in the afternoon (12 p.m. to 4 p.m.), while "t" outcomes slightly increase during business hours (9 a.m. to 4 p.m.) and rebound in the evening. This indicates that users casually browse off-hours and prefer purchasing during the daytime, reflecting greater purchase intent.

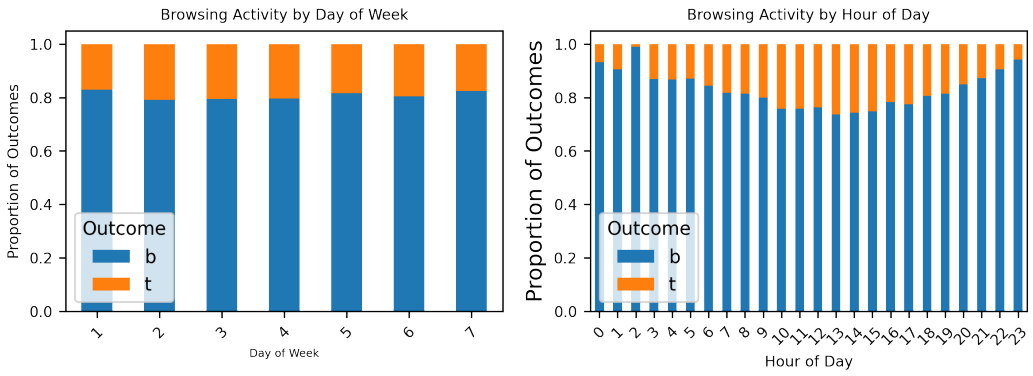


Figure 3. Browsing activity by dow and hod for ds1.

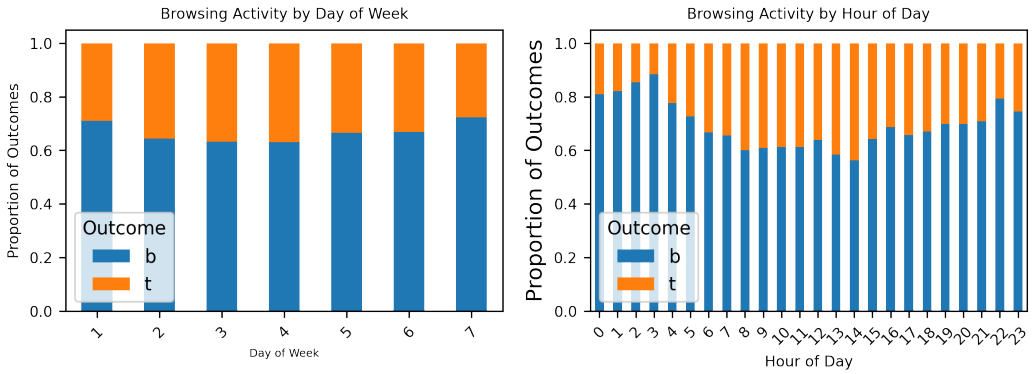


Figure 4. Browsing activity by dow and hod for ds2.

In our subsequent analysis, we assess how session outcomes vary with changes in product prices across datasets (Figures 5 and 6). Both datasets show a right-skewed price distribution, predominantly featuring lower-priced products, with few high-cost items. Purchased products (outcome = 1) generally have slightly lower median prices than non-purchased ones, indicating a preference for less expensive products. Furthermore, both datasets contain significant outliers, especially among non-purchased items, reflecting the presence of costly products that were less likely to sell. This analysis emphasizes pricing's impact on purchasing behavior, with affordability significantly influencing buying decisions.

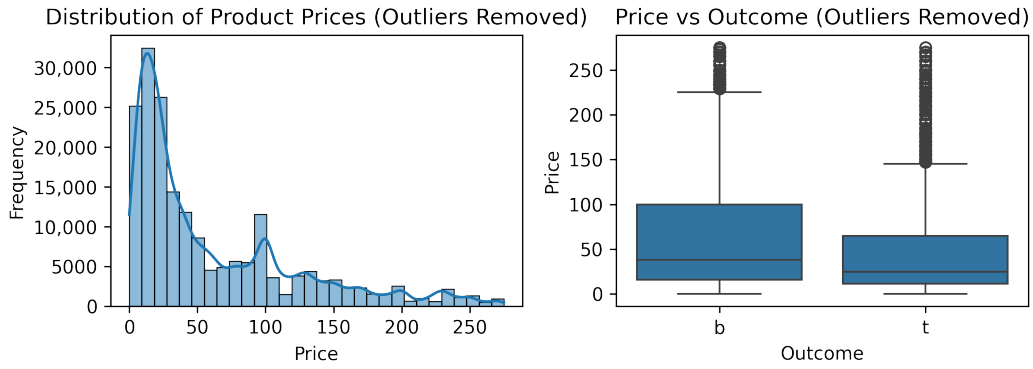


Figure 5. Distribution of product prices (outliers removed) for ds1.

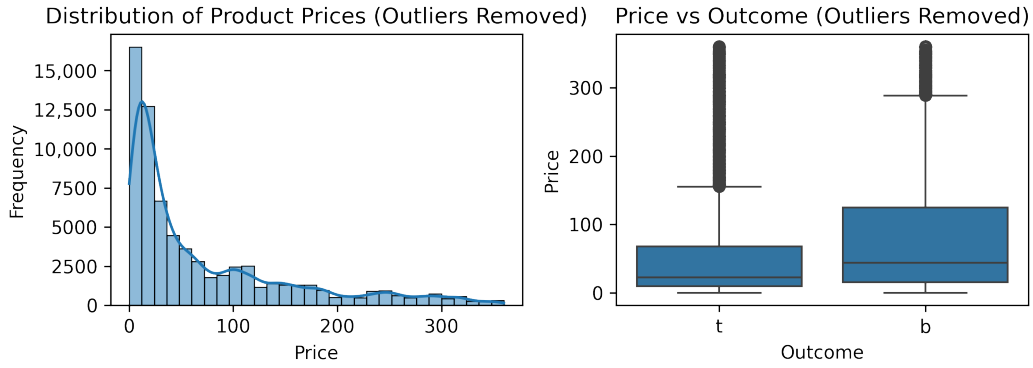
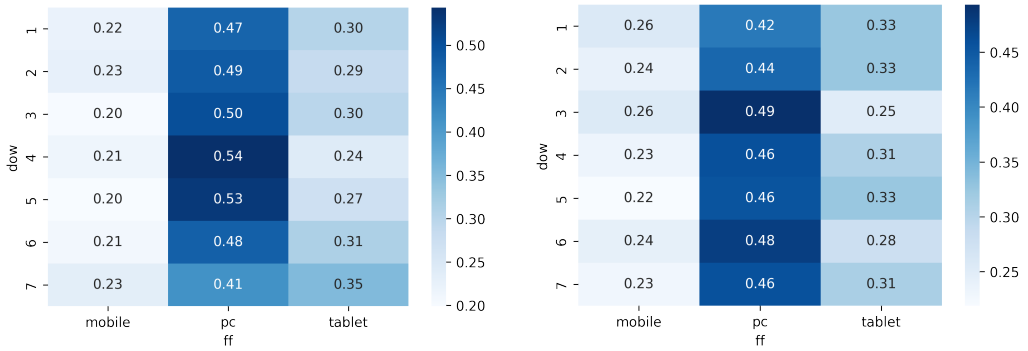


Figure 6. Distribution of product prices (outliers removed) for ds2.

The heatmaps (Figure 7) illustrate the normalized outcome rate by day of week (dow) and device type (ff) for both datasets, revealing consistent patterns across ds1 and ds2. The highest outcome rates are observed midweek, particularly on Wednesdays (dow = 4) and Thursdays (dow = 5), with PC users showing the greatest engagement. For ds1, the

maximum rate reaches 0.54 on Wednesday, while for ds2, it reaches 0.49 on the same day. In contrast, mobile and tablet users demonstrate significantly lower outcome rates throughout the day, generally between 0.20 and 0.33, suggesting that these devices are less conducive to achieving positive results. Weekend days (Saturday and Sunday) exhibit a noticeable decrease in engagement for all device types in both datasets. The consistency in patterns across both datasets highlights the key role of PCs, which consistently outperform other devices in driving results. This suggests that PC users are more likely to take meaningful actions compared to those on mobile or tablet devices. Furthermore, the mid-week peak indicates increased user activity or intent during these days, possibly driven by workweek behaviors.



(**a**) (**b**)

Figure 7. Normalized outcome rate by day of week and device type for ds1 and ds2. (**a**) Normalized outcome rate by day of week (dow) and device type (ff) for ds1; (**b**) normalized outcome rate by day of week (dow) and device type (ff) for ds2.

5. Experiments and Results

To assess the impact of the features extracted from the customer sessions on the performance of the models on cart abandonment prediction and purchase behavior prediction, a series of experiments were carried out. The experimental analysis seeks to address these questions:

1. Which model excels in predicting cart abandonment?
2. How do product category transitions during sessions affect the decision making of prediction models?
3. How do session price dynamics and geographical features influence model decision making?

To explore these questions, experiments are performed with evaluations across datasets and feature importance analysis. Due to the dominance of non-transaction sessions on e-Commerce platforms, datasets are often imbalanced, which can hinder model performance. To mitigate this, we used the synthetic minority oversampling technique (SMOTE) [40] to increase the minority class with synthetic samples. We utilized the grid search method from the sklearn library [41] for optimal parameter selection (Table 2) for traditional ML models, applying these in model training and evaluating using standard metrics. The deep learning model architectures are detailed in Table 3.

Table 2. Search Space for hyperparameter tuning.

Model	Hyperparameters and Values
LR	C: [0.01, 0.1, 1, 10]; penalty: ['l1', 'l2', 'elasticnet']; solver: ['saga']
RF	n_estimators: [50, 100]; max_depth: [5, 10, None]; min_samples_split: [2, 5, 10]
GB	n_estimators: [50, 100]; learning_rate: [0.01, 0.1]; max_depth: [5, 10]
KNN	n_neighbors: [3, 5, 10]; weights: ['uniform', 'distance']
DT	max_depth: [5, 10, 20, None]; min_samples_split: [2, 5, 10]; min_samples_leaf: [1, 2, 5]

Table 3. Deep learning model architectures for cart abandonment prediction.

Model	Architecture
MLP	Dense(128, ReLU) → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.3) → Dense(1, Sigmoid)
CNN	Conv1D(64, ReLU) → MaxPooling → Conv1D(32, ReLU) → Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(1, Sigmoid)
LSTM	LSTM(64) → Dropout(0.3) → LSTM(32) → Dense(64, ReLU) → Dropout(0.3) → Dense(1, Sigmoid)

5.1. Results

These findings underscore significant insights into the models' efficacy in predicting cart abandonment and purchase behaviors across both datasets. In ds1 (Table 4), GB proved superior, attaining the highest F1 score (0.893) and an almost perfect ROC AUC (0.990), indicating its proficiency in balancing false positives and negatives, which is vital for identifying abandoners for precise interventions. Moreover, MLP and RF demonstrated strong performance, with F1 scores of 0.878 and 0.886, highlighting their capacity to manage complex user behavior data. Although CNN and LSTM achieved notable recall values of 0.969 and 0.968, suggesting effectiveness in identifying likely abandoners, their relatively lower precision compared to GB and RF implies a greater likelihood of misclassifying purchasers as abandoners.

Table 4. Model performance comparison for cart abandonment prediction for ds1.

Model	Accuracy	ROC AUC	Precision (t)	Recall (t)	F1-Score (t)
MLP	0.960	0.990	0.811	0.959	0.878
CNN	0.956	0.989	0.789	0.969	0.869
LSTM	0.946	0.987	0.746	0.968	0.843
LR	0.933	0.966	0.710	0.933	0.806
RF	0.963	0.989	0.833	0.947	0.886
GB	0.966	0.990	0.859	0.929	0.893
KNN	0.902	0.954	0.621	0.893	0.733
DT	0.958	0.979	0.808	0.947	0.872

In the case of ds2 (Table 5), RF and GB led the performance rankings, with F1 scores of 0.889 and 0.887, respectively. The gap with MLP closed considerably, since MLP achieved an F1 score of 0.884 and led in precision at 0.836, highlighting its efficacy in reducing false positives in ds2. In particular, the DT model showed improvement over ds1, achieving an F1 score of 0.864 and a recall of 0.966, indicating promise in managing imbalanced cart abandonment issues in ds2. In contrast, LR and KNN consistently underperformed on both datasets, showing limitations in identifying complex cart abandonment patterns.

Table 5. Model performance comparison for cart abandonment prediction for ds2.

Model	Accuracy	ROC AUC	Precision (t)	Recall (t)	F1-Score (t)
MLP	0.911	0.967	0.836	0.937	0.884
CNN	0.905	0.964	0.826	0.935	0.877
LSTM	0.904	0.966	0.812	0.957	0.878
LR	0.842	0.882	0.762	0.819	0.790
RF	0.915	0.967	0.845	0.938	0.889
GB	0.914	0.969	0.852	0.924	0.887
KNN	0.778	0.859	0.657	0.814	0.727
DT	0.890	0.959	0.781	0.966	0.864

In cross-dataset comparisons (Tables 4 and 5), the models exhibited superior performance in ds1, achieving higher accuracy and F1 scores. This indicates that ds1 features more distinct or predictable patterns of abandonment of the cart and purchase behaviors. Neural networks (MLP, CNN, LSTM) showed versatility with both datasets, with MLP achieving a stable precision–recall trade-off. Ensemble methods such as GB and RF consistently excelled, marking them as prime candidates for actionable cart abandonment predictions. In conclusion, GB is the preferred option for predicting purchase behavior after products are added to the cart, owing to its superior precision and recall, with RF and MLP also serving as viable alternatives.

5.2. Insights on Factors Influencing Cart Abandonment

The permutation importance for the GB model reveals valuable insights on factors influencing cart abandonment in Table 6. For ds1, the primary factor influencing purchase behavior is the number of previous purchases, with an importance score of 0.827. The next most important factor is the number of pages viewed at an importance score of 0.070, followed by the number of previous sessions with 0.050. Timing elements such as hod (hour of the day) and dom (day of the month) have lesser significance, scoring 0.011 and 0.010, respectively, suggesting their limited effect compared to browsing and buying patterns.

For ds2, the results suggest that the number of previous purchases (importance: 0.684) exerts a significant influence, though to a lesser degree than in ds1. The number of previous sessions (importance: 0.080) is noteworthy as well. Moreover, the number of pages viewed (importance: 0.061) remains a critical factor, aligning with the ds1 pattern. Longitude and latitude, which were less crucial in ds1, show increased importance in ds2 (0.025 and 0.022), implying possible geographic influences or varying user behaviors by location. Features like hod and dow, while still relevant, hold lower importance compared to other features, with hod at 0.015 and dow at 0.008. Additionally, min_price (0.022) and max_price (0.021) are significant, indicating that price ranges impact purchase or cart abandonment more in this dataset.

We investigate factors affecting prediction model accuracy through SHAP analysis of the datasets in Figure 8. The analysis shows that consumer loyalty and dynamic factors like price changes and product category shifts significantly impact predictions on purchases or cart abandonment. Key predictors include the number of previous purchases, hour (hour of the day), and number of previous sessions in both datasets. Dynamic pricing factors, such as min price and max price, are notably influential, especially in ds2. Conversely, price change and price difference are less impactful, implying that while price shifts affect consumer behavior, they are less critical to predictions. The feature cat changes (product category changes) is prominent, highlighting that category shifts during sessions have a stronger effect on abandonment predictions than price alone. Thus, businesses should focus on past user behaviors and session dynamics, such as category changes, to enhance abandonment

prediction models. Although price changes remain relevant, their impact is secondary to immediate session activities and product interactions in guiding purchase decisions.

Table 6. Permutation importance for GB in cart abandonment prediction.

Feature	ds1 Importance	ds2 Importance
number_of_previous_purchases	0.827	0.684
number_of_pages_viewed	0.070	0.061
number_of_previous_sessions	0.050	0.080
hod	0.011	0.015
dom	0.010	0.014
dow	0.008	0.008
products_viewed	0.007	0.006
cat_changes	0.004	0.013
max_price	0.003	0.021
longitude	0.003	0.025
min_price	0.002	0.022
unique_prices	0.002	0.003
latitude	0.002	0.022
price_difference	0.001	0.012
ff_pc	0.000	0.011
ff_tablet	0.000	0.002
weekend	0.000	0.001
price_change	0.000	0.000

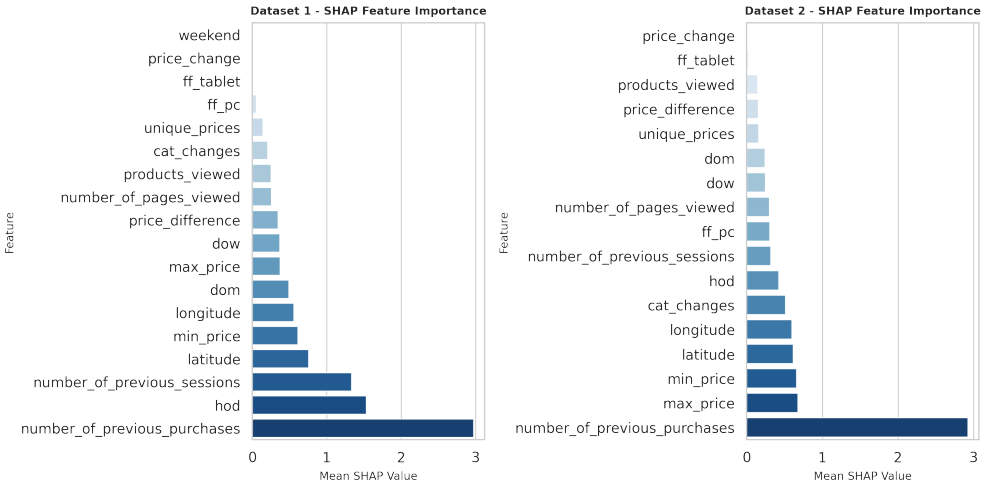


Figure 8. SHAP feature importance analysis for predicting cart abandonment and purchase behavior.

Drawing from the results of our experimental analysis, we revisit our research questions as follows:

RQ1: The transitions of the feature category (cat changes) moderately influence both datasets. Despite the low importance of the permutation (0.004 in ds1, 0.013 in ds2), the SHAP value notably increases in ds2 (0.512 from 0.208 in ds1), indicating that these transitions considerably impact the model output in ds2. This suggests that users who navigate multiple categories might have different decision patterns. However, their impact is secondary to dominant behavior traits like number of previous purchases (SHAP: 2.9–3.0), showing that category shifts are relevant but not central to model predictions.

RQ2: The volatility of the price and the extremes (max price, min price, price difference) exhibit a stronger influence in ds2: Permutation importance (max price) (0.021 in ds2 vs. 0.003 in ds1), min price (0.022 in ds2 vs. 0.002 in ds1) and (max price) has the second highest magnitude in ds2 (0.679 in ds2 vs. 0.372 in ds1). This implies that price dynamics

play a larger role in ds2, potentially capturing sensitivity to high/low prices or price fluctuations. For example, extreme prices might be correlated with hesitation (abandonment) or deliberate purchase intent, but SHAP magnitudes alone cannot confirm directionality.

RQ3: Geographic features (latitude, longitude) have more influence in ds2, with permutation importance around 0.022–0.025 versus 0.002 in ds1. The SHAP values denote a moderate influence of geographic variables on both datasets (latitude: 0.755 in ds1, 0.616 in ds2), which implies the existence of regional trends or localized patterns in user behavior.

6. Discussion

This study investigates predicting purchase behavior following cart addition utilizing eight distinct machine learning models on two real-world datasets. The results indicate that GB consistently outperforms the other models, achieving the highest accuracy, ROC AUC, precision, recall, and F1 score in both datasets. Specifically, GB achieved superior performance in terms of prediction accuracy and balanced metrics, demonstrating its robustness in identifying patterns related to purchase prediction following cart addition. Additionally, RF also delivered strong results, outperforming most models except GB, while deep learning models such as MLP and LSTM showed varied performance depending on the dataset. This highlights the importance of selecting the appropriate model based on the characteristics and complexity of the dataset, as these factors significantly affect prediction performance.

Examining the importance of the features via SHAP values (Figure 8) reveals consistent trends across datasets. In both, the count of prior purchases is the most influential factor, notably in ds1, where a SHAP value of 2.97 underscores its strong predictive ability for purchase behavior. The pages viewed are also crucial, with SHAP values of 0.25 for ds1 and 0.30 for ds2, indicating that more browsing correlates with more accurate purchase behavior prediction following cart addition. The number of previous sessions significantly affects ds2 (SHAP value of 0.32), suggesting that frequent visits are better predictors of purchase behavior. The key differences between the datasets include the decrease in the significance of hours of the day from 0.58 in ds1 to 0.43 in ds2 and in the day of the week from 0.37 to 0.25. In contrast, the importance of geographic features such as latitude and longitude increases in ds2 (0.62 and 0.60) relative to ds1 (0.76 and 0.56), probably reflecting varying user behaviors or market segments per dataset.

Another crucial predictor is category change, particularly in ds2 (SHAP value of 0.51), highlighting the impact of shifting product categories on accurate purchase prediction following cart addition. This finding underscores the significance of users' reactions to changing product offerings in predicting purchase or abandonment behaviors. In ds1, the influence of the category change is smaller (SHAP value of 0.21) but still significant, indicating that the impact varies with the characteristics of the dataset or patterns of consumer behavior. Price-related features, notably the minimum and maximum price, exert a stronger influence on ds2, with SHAP values of 0.61 and 0.37 in ds1, and 0.66 and 0.68 in ds2, suggesting greater price sensitivity in ds2.

Our findings align with key studies on purchase behavior. For example, Kim et al. [24] linked browsing patterns to purchase intent, and our SHAP values for number of pages viewed (ds1: 0.25, ds2: 0.30) and number of previous sessions(ds2: 0.32) reflect this influence. Similarly, the importance of the feature of the history of prior purchases was emphasized in [32], validating its dominance in our models (ds1: 2.97, ds2: 2.92). Correspondingly, Chaudhuri et al. [21]'s focus on price sensitivity aligns with our min price results (ds1: 0.61, ds2: 0.66). These consistencies support user engagement and price anchoring as universal predictors in both datasets (ds1, ds2) and methodologies. To our knowledge, no previous work has explicitly examined the impact of category changes

during sessions (e.g., cat changes, SHAP: ds1: 0.21, ds2: 0.51) or geographical features (latitude, longitude, SHAP: ds1: 0.76/0.56, ds2: 0.62/0.60) on the prediction decision of ML models for cart abandonment or purchase.

6.1. Contributions

This study addresses the gaps present in the session-level analysis by incorporating features such as product category transitions, price dynamics, and geographic attributes into predictive models for cart abandonment and purchase behavior following cart addition. This integration improves our understanding of consumer decision-making processes and provides actionable insights for e-commerce, including dynamic pricing and region-specific strategies to increase conversion rates, while establishing a foundation for future advancements in e-commerce analytics. This is achieved by initially incorporating the often-overlooked product category dynamics, which, although not the most predictive, provide context regarding consumer interests and exploratory behavior. Furthermore, the examination of price dynamics, previously neglected, shows that minimum and maximum prices significantly influence purchase behavior, particularly in the ds2 dataset, thereby underscoring the importance of price sensitivity.

Geographic attributes contribute an additional contextual layer to the models by aligning consumer behavior with regional factors, thus enriching the models by acknowledging the underrepresented influence of context on consumer behavior, particularly in ds2. Furthermore, the SHAP-based analysis uncovers dataset-specific patterns, revealing that prior purchases are predominant in ds1, while price dynamics and geographic attributes are crucial in ds2. This validates the importance of dynamic session-level features as critical predictors and underscores the necessity for adaptive modeling strategies.

6.1.1. Theoretical Implications

This research sheds light on the theoretical framework of online consumer behavior, focusing on predicting purchase actions after product cart additions. Unlike earlier work [1,15,24,32,42] that emphasized consumer loyalty, historical purchases, and browsing patterns, this study incorporates often neglected elements like session-level price dynamics, geographic data, and category changes. By emphasizing dynamic and situational factors, this research strengthens the model of consumer decision making in online shopping for purchase prediction post-cart addition. Notably, it introduces price-related features—such as session minimum and maximum prices, and price variations—which reveal new perspectives on how real-time price shifts affect consumer behavior. This departs from prior research treating price as static [43], highlighting the role of intrasession price variability in shaping purchase decisions after cart additions, thus challenging conventional pricing models.

Incorporating geographic indicators, such as latitude and longitude, brings a new dimension to the prediction of online purchases. Geographic segmentation is well studied in physical retail, yet its e-commerce applications remain limited. This study demonstrates that location influences purchase predictions after cart addition, indicating that regional consumer behavior, logistics, and market dynamics should be included in online shopping models. Furthermore, it provides new information on consumer exploration by examining changes in the category in the session, an aspect previously underexplored in purchase decisions after cart addition [44,45]. By recognizing category changes as insightful predictors, we enhance our theoretical understanding of consumer uncertainty and the exploration affecting decisions. Our findings also extend decision-making theories by highlighting how habitual behaviors, evidenced by past purchases and sessions, interact with session-specific factors such as pricing and category changes. The models confirm habitual actions as a key

factor [32,42], but they interact with dynamic session variables, advocating for a detailed theoretical framework to grasp online consumer behavior.

6.1.2. Practical Implications

This study offers crucial practical insights for businesses seeking to enhance the accuracy of cart abandonment and purchase forecasts. While conventional models have centered on factors like consumer loyalty [32] and browsing habits [12,24], our research contributes meaningfully by adding dynamic elements such as price variations and session-specific product category shifts, particularly focusing on sessions with cart additions. These often-overlooked elements are important for precise predictions of session outcomes.

Our feature importance analysis demonstrates that incorporating price dynamics and the frequency of changes in product categories, in addition to consumer loyalty metrics, is crucial for understanding consumer purchase behavior once products are added to the cart. This highlights that businesses cannot depend solely on conventional session features to predict purchase decisions. In order to improve purchase forecasts after cart additions, businesses should incorporate real-time modifications considering sessional price fluctuations and product category shifts, as well as consumer purchase history, into their ML models.

Businesses can employ strategies like personalized discounts or dynamic product displays to anticipate potential abandonment. This can also facilitate real-time marketing actions, such as personalized notifications or targeted promotions tailored to session behavior, enhancing conversion rates. By incorporating these dynamic features into ML models, companies can not only increase the accuracy of predicting purchase behaviors once items are added to the cart, but also enhance the overall customer experience through more individualized and responsive approaches.

7. Conclusions

This study investigates factors impacting cart abandonment and purchase decisions in online shopping following cart addition, focusing on evaluating machine learning models for predictive accuracy. Among the eight models tested, GB proved the best at identifying patterns in purchase behavior. Key influencing factors included historical purchase activity, session frequency, and pages viewed, linked to user engagement, while geographical factors such as location indicated regional preferences. Price sensitivity was crucial, with little impact from price trends and moderate effects from category changes. This research underscores the importance of predictive models that integrate dynamic elements such as historical behavior, session details, pricing, and location.

Our findings provide insights to reduce cart abandonment and increase conversions. Platforms can use historical purchase data, such as past transactions and session frequency, to personalize retargeting for high-risk users with incentives like discounts or rewards. Geographic differences in abandonment rates offer opportunities to refine regional pricing, shipping, or inventory to fit local preferences. For price-sensitive users, real-time strategies like price-match guarantees or flash sales can reduce hesitation. Session metrics, such as pages viewed or category shifts, suggest ways to simplify navigation or recommend complementary products, minimizing decision fatigue. Integrating these insights into predictive models like GB can help e-Commerce platforms address abandonment and improve user experience through personalization.

This study has several limitations. First, the analysis relied on two datasets from a single SaaS provider, limiting generalizability. Future work should incorporate domain-specific datasets, for instance, contrasting sectors like luxury goods and fast fashion or subscription and one-time purchase models, alongside emerging market data from regions

such as Southeast Asia and Africa to evaluate how cultural, economic, or platform-specific factors influence feature importance. Second, while our models focused on session-level and geographic features, expanding to detailed user behavior data, such as cursor movements, wishlist interactions, or responsiveness to real-time promotions such as flash sale click-through rates, could refine prediction accuracy. Third, integrating external data such as social media sentiment, for example, brand mentions on Twitter/X, or supply-chain disruptions, including stockouts, may uncover novel abandonment triggers. Finally, to mitigate privacy risks with sensitive data such as latitude/longitude, researchers could adopt federated learning frameworks to preserve user anonymity without sacrificing analytical depth.

Author Contributions: Conceptualization, R.E.; Formal Analysis, A.G.; Investigation, A.G.; Methodology, R.E. and A.G.; Validation, R.E.; Writing—Original Draft, R.E.; Writing—Review and Editing, A.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author due to privacy concerns.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Khandokar, I.A.; Islam, A.M.; Islam, S.; Shatabda, S. A gradient boosting classifier for purchase intention prediction of online shoppers. *Heliyon* **2023**, *9*, e15163. [CrossRef]
2. Chaubey, G.; Gavhane, P.R.; Bisen, D.; Arjaria, S.K. Customer purchasing behavior prediction using machine learning classification techniques. *J. Ambient. Intell. Humaniz. Comput.* **2023**, *14*, 16133–16157. [CrossRef]
3. Kapoor, A.P.; Vij, M. Following you wherever you go: Mobile shopping 'cart-checkout' abandonment. *J. Retail. Consum. Serv.* **2021**, *61*, 102553. [CrossRef]
4. Wang, S.; Ye, Y.; Ning, B.; Cheah, J.H.; Lim, X.J. Why do some consumers still prefer in-store shopping? An exploration of online shopping cart abandonment behavior. *Front. Psychol.* **2022**, *12*, 829696. [CrossRef]
5. Wang, S.; Cheah, J.H.; Lim, X.J. Online shopping cart abandonment: A review and research agenda. *Int. J. Consum. Stud.* **2023**, *47*, 453–473. [CrossRef]
6. Koehn, D.; Lessmann, S.; Schaal, M. Predicting online shopping behaviour from clickstream data using deep learning. *Expert Syst. Appl.* **2020**, *150*, 113342. [CrossRef]
7. Jannach, D.; Jugovac, M. Measuring the business value of recommender systems. *ACM Trans. Manag. Inf. Syst. (TMIS)* **2019**, *10*, 1–23. [CrossRef]
8. Luo, X.; Lu, X.; Li, J. When and how to leverage e-commerce cart targeting: The relative and moderated effects of scarcity and price incentives with a two-stage field experiment and causal forest optimization. *Inf. Syst. Res.* **2019**, *30*, 1203–1227. [CrossRef]
9. Wang, S.; Lim, X.J.; Luo, X.; Cheah, J.H. To hesitate or not to hesitate: Can popularity cues minimize the hesitation to checkout in e-commerce? *J. Retail. Consum. Serv.* **2024**, *78*, 103730. [CrossRef]
10. Hashemian, F.; Maleki, N.; Zeinali, Y. From User Behavior to Subscription Sales: An Insight into E-Book Platform Leveraging Customer Segmentation and A/B Testing. *Serv. Mark. Q.* **2024**, *45*, 153–181. [CrossRef]
11. Campbell, C.; Sands, S.; Ferraro, C.; Tsao, H.Y.J.; Mavrommatis, A. From data to action: How marketers can leverage AI. *Bus. Horizons* **2020**, *63*, 227–243. [CrossRef]
12. Rausch, T.M.; Derra, N.D.; Wolf, L. Predicting online shopping cart abandonment with machine learning approaches. *Int. J. Mark. Res.* **2022**, *64*, 89–112. [CrossRef]
13. Kukar-Kinney, M.; Close, A.G. The determinants of consumers' online shopping cart abandonment. *J. Acad. Mark. Sci.* **2010**, *38*, 240–250. [CrossRef]
14. Rubin, D.; Martins, C.; Ilyuk, V.; Hildebrand, D. Online shopping cart abandonment: A consumer mindset perspective. *J. Consum. Mark.* **2020**, *37*, 487–499. [CrossRef]
15. Padigela, P.K.; Suguna, R. Segmentation of E-commerce users based on cart abandonment and product recommendation through collaborative filtering: The moderating effect of exorbitant pricing. *Int. J. Syst. Assur. Eng. Manag.* **2021**, 1–10. [CrossRef]

16. Jiang, D.; Zhang, G.; Wang, L. Empty the shopping cart? The effect of shopping cart item sorting on online shopping cart abandonment behavior. *J. Theor. Appl. Electron. Commer. Res.* **2021**, *16*, 1973–1996. [CrossRef]
17. Zhao, H.; Wang, X.; Jiang, L. To purchase or to remove? Online shopping cart warning pop-up messages can polarize liking and purchase intention. *J. Bus. Res.* **2021**, *132*, 813–836. [CrossRef]
18. Li, C.; Chen, J.E.; Peng, S.; Huang, J.; Sha, X. Examining the effects of weather on online shopping cart abandonment: Evidence from an online retailing platform. *J. Retail. Consum. Serv.* **2024**, *78*, 103738. [CrossRef]
19. Rodrigues, M.A.; Proença, J.; Soares, I. The influence of perceived risk on mobile shopping cart abandonment. *J. Strateg. Mark.* **2024**, *32*, 152–169. [CrossRef]
20. Wang, S.; Cheah, J.H.; Lim, X.J.; Leong, Y.C.; Choo, W.C. Thanks COVID-19, I'll reconsider my purchase: Can fear appeal reduce online shopping cart abandonment? *J. Retail. Consum. Serv.* **2022**, *64*, 102843. [CrossRef]
21. Chaudhuri, N.; Gupta, G.; Vamsi, V.; Bose, I. On the platform but will they buy? Predicting customers' purchase behavior using deep learning. *Decis. Support Syst.* **2021**, *149*, 113622. [CrossRef]
22. Xiao, F.; Chen, S.X.; Chen, Z.Y.; Wang, Y.N.; Wang, J.Q. An interpretable data-driven approach for customer purchase prediction using cost-sensitive learning. *Eng. Appl. Artif. Intell.* **2024**, *138*, 109344. [CrossRef]
23. Ehsani, F.; Hosseini, M. Customer purchase prediction in electronic markets from clickstream data using the Oracle meta-classifier. *Oper. Res.* **2024**, *24*, 11. [CrossRef]
24. Kim, S.; Shin, W.; Kim, H.W. Predicting online customer purchase: The integration of customer characteristics and browsing patterns. *Decis. Support Syst.* **2024**, *177*, 114105. [CrossRef]
25. Trivedi, S.K.; Patra, P.; Srivastava, P.R.; Zhang, J.Z.; Zheng, L.J. What prompts consumers to purchase online? A machine learning approach. *Electron. Commer. Res.* **2022**, *24*, 2953–2989. [CrossRef]
26. Wei, Z.; Lihua, C.; Lijuan, X. Prediction Model for E-Commerce User Purchase Behavior Based on Data Mining. In Proceedings of the 2024 IEEE 4th International Conference on Electronic Communications, Internet of Things and Big Data (ICEIB), Taiwan, China, 19–21 April 2024 ; pp. 583–586.
27. Wu, Z.; Chen, X.; Gao, Z. Bayesian non-parametric method for decision support: Forecasting online product sales. *Decis. Support Syst.* **2023**, *174*, 114019. [CrossRef]
28. Liu, Z.; Zhang, Y.; Abedin, M.Z.; Wang, J.; Yang, H.; Gao, Y.; Chen, Y. Profit-driven fusion framework based on bagging and boosting classifiers for potential purchaser prediction. *J. Retail. Consum. Serv.* **2024**, *79*, 103854. [CrossRef]
29. Chen, S.; Ngai, E.W.; Xiao, F.; Xu, Z. From comparison to purchasing: Effects of online behavior toward associated co-visited products on consumer purchase. *Inf. Manag.* **2024**, *61*, 103938. [CrossRef]
30. Mikalef, P.; Sharma, K.; Chatterjee, S.; Chaudhuri, R.; Parida, V.; Gupta, S. All eyes on me: Predicting consumer intentions on social commerce platforms using eye-tracking data and ensemble learning. *Decis. Support Syst.* **2023**, *175*, 114039. [CrossRef]
31. Sakthi, B.; Sundar, D. An efficient attention-based hybridized deep learning network with deep RBM features for customer behavior prediction in digital marketing. *Kybernetes* **2024**. [CrossRef]
32. Esmeli, R.; Bader-El-Den, M.; Abdullahi, H. An analyses of the effect of using contextual and loyalty features on early purchase prediction of shoppers in e-commerce domain. *J. Bus. Res.* **2022**, *147*, 420–434. [CrossRef]
33. Sakar, C.O.; Polat, S.O.; Katircioglu, M.; Kastro, Y. Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks. *Neural Comput. Appl.* **2019**, *31*, 6893–6908. [CrossRef]
34. Kotler, P.; Keller, K.L. *Marketing Management*, 15th ed.; Pearson: Harlow, UK, 2016.
35. Boateng, E.Y.; Abaye, D.A. A review of the logistic regression model with emphasis on medical research. *J. Data Anal. Inf. Process.* **2019**, *7*, 190. [CrossRef]
36. Belgiu, M.; Drăguţ, L. Random forest in remote sensing: A review of applications and future directions. *ISPRS J. Photogramm. Remote Sens.* **2016**, *114*, 24–31. [CrossRef]
37. Bentéjac, C.; Csörgő, A.; Martínez-Muñoz, G. A comparative analysis of gradient boosting algorithms. *Artif. Intell. Rev.* **2021**, *54*, 1937–1967. [CrossRef]
38. Mucherino, A.; Papajorgji, P.J.; Pardalos, P.M.; Mucherino, A.; Papajorgji, P.J.; Pardalos, P.M. K-nearest neighbor classification. *Data Min. Agric.* **2009**, *34* , 83–106.
39. Pan, H.; Zhou, H. Study on convolutional neural network and its application in data mining and sales forecasting for E-commerce. *Electron. Commer. Res.* **2020**, *20*, 297–320. [CrossRef]
40. Chawla, N.V.; Bowyer, K.W.; Hall, L.O.; Kegelmeyer, W.P. SMOTE: synthetic minority over-sampling technique. *J. Artif. Intell. Res.* **2002**, *16*, 321–357. [CrossRef]
41. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; et al. Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* **2011**, *12*, 2825–2830.
42. Martínez, A.; Schmuck, C.; Pereverzyev, S., Jr.; Pirker, C.; Haltmeier, M. A machine learning framework for customer purchase prediction in the non-contractual setting. *Eur. J. Oper. Res.* **2020**, *281*, 588–596. [CrossRef]

43. Satriawan, K.A.; Setiawan, P.Y. The role of purchase intention in mediating the effect of perceived price and perceived quality on purchase decision. *Int. Res. J. Manag. Soc. Sci.* **2020**, *7*, 38–49. [CrossRef]
44. Dong, X.; Tu, H.; Zhu, H.; Liu, T.; Zhao, X.; Xie, K. Does diversity facilitate consumer decisions: A comparative perspective based on single-category versus multi-category products. *Asia Pac. J. Mark. Logist.* **2024**, *36*, 936–956. [CrossRef]
45. Dong, X.; Li, B.; Xie, K.; Zhao, X.; Xiao, R. The effect of the product categories diversity recommended on cross-buying in electronic commerce platforms: The moderating role of user navigation heterogeneity. *J. Consum. Behav.* **2024**, *24*, 304–331. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.